

Agent-based modeling of firm's heterogeneous preferences: Implications for trading and technology adoption in electricity-carbon markets

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ARTICLE INFO

Keywords:

Agent-based model
Multi-agent reinforcement learning
Carbon trading
Electricity market
Heterogeneous preferences

ABSTRACT

Firm preference and behavior significantly influence carbon market performance and emission reduction efficiency. This paper develops an electricity-carbon coupled market model, integrating Agent-Based Modeling (ABM) and Multi-Agent Reinforcement Learning (MARL), to analyze the trading and investment behavior of heterogeneous firms in carbon markets. Using calibrated data from China's energy markets, the study uncovers four key findings. First, “compliance-based trading” behavior drives carbon price and volume surges near compliance deadlines. Second, large firms, with financial and technical advantages, act as first movers in adopting low-carbon technologies. Third, longer payback period reduces carbon prices but promote technological innovation, while stronger technology-oriented expectations boost trading activity, prices, and innovation. Fourth, trading preferences shape market outcomes: arbitrage firms increase short-term price volatility, risk-hedge firms stabilize markets and lead in innovation, and speculative firms strike a balance between price impacts and moderate innovation. Policy recommendations include extending payback period to ease financial pressures and encourage technology diffusion, leveraging large firms' resources while supporting smaller firms with fiscal incentives, and regulating arbitrage behavior during compliance periods to stabilize carbon markets.

1. Introduction

The emission trading system (ETS) has emerged as a vital market-based policy instrument for addressing climate change by reducing greenhouse gas emissions and incentivizing low-carbon technological innovation (Newell et al., 2014; Flora and Variolo, 2020; Zhou et al., 2022a). By setting emissions caps and enabling the trading of carbon allowances, ETSs seek to align economic incentives with environmental objectives, fostering cost-effective emissions reductions across industries. However, the effectiveness of ETSs in achieving these goals varies significantly across regions and sectors, influenced by market dynamics, firm-level behavior, and regulatory frameworks (Naegle and Zaklan, 2019; Nazifi et al., 2021; Wang et al., 2023). Understanding the mechanisms driving these variations is essential for improving market performance and ensuring the long-term sustainability of carbon trading systems.

Firm-level preferences and decision-making behavior play a pivotal role in determining the market dynamics and ETS efficiency (Thema et al., 2013; Ghosh et al., 2015). These preferences are diverse and

multifaceted, encompassing investment horizons, technological priorities, and trading strategies (Palao and Pardo, 2017; Hao et al., 2023). For example, firms with short payback period expectations may favor low-cost, compliance-driven strategies such as purchasing carbon allowances, while those with a focus on long-term competitiveness might prioritize investments in advanced low-carbon technologies. Similarly, trading preferences—such as speculative, arbitrage, or risk-hedge strategies—affect market liquidity, price stability, and the adoption and diffusion of low-carbon technologies (Yu and Chen, 2018; Yao and Alexiou, 2022). Firms motivated by risk-hedging preferences may accelerate technology adoption to mitigate regulatory and price uncertainties, whereas speculative or arbitrage-driven firms might exhibit a slower, more opportunistic approach. Such behavioral diversity underscores the need for analytical frameworks that capture the interactions between market trading and technological innovation (Zhou et al., 2022b; Yoon et al., 2024).

Agent-Based Modeling (ABM) provides a robust methodology for examining the complex dynamics of ETSs, particularly in markets characterized by heterogeneous agents and bounded rationality

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<https://doi.org/10.1016/j.eneeco.2025.108688>

Received 30 December 2024; Received in revised form 22 May 2025; Accepted 19 June 2025

Available online 24 June 2025

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(Richstein et al., 2014; Wei et al., 2023; Wu et al., 2024). By simulating the behavior and interaction of individual firms, ABM enables a granular analysis of how diverse preferences influence market outcomes and technology adoption (Hoekstra et al., 2017; Araavind et al., 2024; Torralba-Díaz et al., 2024). Existing studies have leveraged ABM to explore various aspects of carbon markets, including trading mechanisms, policy impacts, and technological transitions, in industries such as power generation, steel production, and petrochemicals (Ringler et al., 2016; Yu et al., 2020; Fang and Ma, 2020).

Recent advancements in integrating ABM with Multi-agent Reinforcement Learning (MARL) further enhance its ability to model adaptive behavior in dynamic and high-dimensional environments (Rahimiyan and Mashhadi, 2010; Liu et al., 2021; Harrold et al., 2022). For instance, Sun et al. (2022) developed a multi-energy market framework, where an enhanced MADDPG algorithm was employed to coordinate energy management among heterogeneous agents under uncertain market conditions. Their work highlights how dual mechanisms of energy and carbon trading can guide firms towards optimal consumption patterns while ensuring transaction fairness and data privacy. Notably, the improved MADDPG algorithm achieved superior convergence and stability compared to traditional methods, offering a robust solution for multi-layered market coordination. Similarly, Wu et al. (2022) proposed a competitive electricity market model using MARL with transfer learning, where Q-learning-based bidding strategies enabled power plants to dynamically adapt to market clearing prices. Their results revealed that MARL-driven strategies outperform rule-based approaches in highly diversified and competitive environments, particularly in scenarios with asymmetric information and strategic interactions. While these studies underscore the value of MARL in simulating energy markets, critical gaps remain. First, existing frameworks often oversimplify agent heterogeneity, such as the technical and behavioral diversity observed in real-world firms. Second, prior work predominantly focuses on electricity or gas markets, with limited attention to the unique dynamics of carbon markets, such as compliance-driven trading and allowance allocation mechanisms.

Building on these foundations, this paper aims to explore the interconnections between firm-level trading behavior, low-carbon technology adoption, and the performance of ETSS. Using China's carbon market as a case study, this research develops a comprehensive electricity-carbon coupled market model with the following innovations. First, it incorporates a heterogeneous firm framework that captures diverse decision-making attributes, including investment preferences, technological priorities, and trading strategies, to analyze their impacts on market outcomes. Second, a multi-level market structure is introduced to simulate interactions across day-ahead, intraday, and medium-to-long-term markets, offering a holistic perspective on the interplay between short-term trading dynamics and long-term technology adoption. Third, the model employs an advanced MADDPG algorithm to enhance the decision-making and adaptive capabilities of these heterogeneous agents within high-dimensional and dynamic settings. This specifically extends the MADDPG algorithm with attention mechanisms and multi-timescale learning, enabling the agents to better capture the temporal dependencies and strategic complexities inherent in carbon-electricity coupled systems, thereby advancing beyond the static or simplified environments in previous studies.

This study contributes to the broader literature by providing insights into how firm-level heterogeneity shapes the functionality and effectiveness of ETSS. The findings have significant implications for both policymakers and market participants, offering guidance on designing carbon markets that balance short-term efficiency with long-term sustainability. The remainder of the paper is organized as follows. Section 2 outlines the theoretical framework and methodological approach. Section 3 presents the simulation results and discusses their implications. Finally, Section 4 concludes with key insights and recommendations for future research and practice.

2. The electricity-carbon coupled market trading mechanism model

The model introduced a multi-timescale framework to reflect the influence of incomplete information on corporate behavior and the market changes caused by irrational corporate decisions as relevant studies (Yu et al., 2020; Wei et al., 2022). We simulate the electricity-carbon coupled market trading mechanism that lasts for T periods in which each period is denoted by t , which represents one "day". The carbon market is further divided into several "seconds" in each "day", representing the individual transactions under the continuous two-way auction mechanism.

The proposed electricity-carbon coupled market trading mechanism model consists of four modules: firm module, electricity market module, government module, and carbon market module. Within this framework, the joint decision-making process of electricity firms participating in both carbon and electricity markets is modeled as a multi-period decision process, encompassing long-term trading decisions, low-carbon technology investment decisions, output decisions, and emissions allowance trading decisions. The decision-making flowchart is illustrated in Fig. 1.

2.1. Firm module

Building upon scale, cost, and risk preference, we further introduce multi-dimensional static attributes such as technological levels and information endowments to characterize the heterogeneity among firm agents, as Chen and Ma (2021) illustrated. The model defines the profit of the firm i as π_i . $R_{i,t}$ and $C_{i,t}$ represent the revenue and production cost of the firm in period t . Each firm i attempts to maximize the profit with the following objective function:

$$\max \pi_i = \sum_{t=1}^T (R_{i,t} - C_{i,t}) \quad (1)$$

Electricity firms primarily obtain revenue by participating in the spot market and the medium- to long-term market. $R_{i,t}^{DA}$, $R_{i,t}^{ID}$, $R_{i,t}^M$, $R_{i,t}^Y$ denotes the revenue from electricity sales of firms participating in the day-ahead, intra-day, monthly and annual markets; p_t^{DA} and p_t^{ID} represents the spot market electricity prices for day-ahead and intraday periods, p_t^M and p_t^Y represents the long-term contract electricity prices for monthly and annual periods. $q_{i,t}^{DA}$, $q_{i,t}^{ID}$, $q_{i,t}^M$, and $q_{i,t}^Y$ represents the trading volume in different markets. LL_t is line loss rate, meaning the loss problem in the transmission process from power generation to grid connection. The revenue of firm i is computed as follows:

$$\begin{cases} R_{i,t} = R_{i,t}^{DA} + R_{i,t}^{ID} + R_{i,t}^M + R_{i,t}^Y \\ R_{i,t}^{DA} = p_t^{DA} \times q_{i,t}^{DA} \times (1 - LL_t) \\ R_{i,t}^{ID} = p_t^{ID} \times q_{i,t}^{ID} \times (1 - LL_t) \\ R_{i,t}^M = p_t^M \times q_{i,t}^M \times T^M \\ R_{i,t}^Y = p_t^Y \times q_{i,t}^Y \times T^Y \end{cases} \quad (2)$$

The total cost of each firm comprises operation and maintenance (O&M) costs ($C_{i,t}^{OM}$), fuel costs ($C_{i,t}^{fuel}$), carbon trading costs ($C_{i,t}^{em}$), and technology adoption costs ($C_{i,t}^{Act}$). $C_{i,t}^{OM}$ is related to the unit capacity O&M cost μ_i multiplied by the total installed capacity IC_i ; $C_{i,t}^{fuel}$ is obtained by multiplying the fuel consumption ($q_{i,t}^f$) and the fuel price (p^f); $p_{i,k}^A$ is the trading price in carbon market, and $x_{i,t,k}$ is the trading volume of the firm in carbon market; CI_j denotes the investment cost of technology j , PB_i is the payback period of firm i . The total cost of the enterprise is calculated as follows:

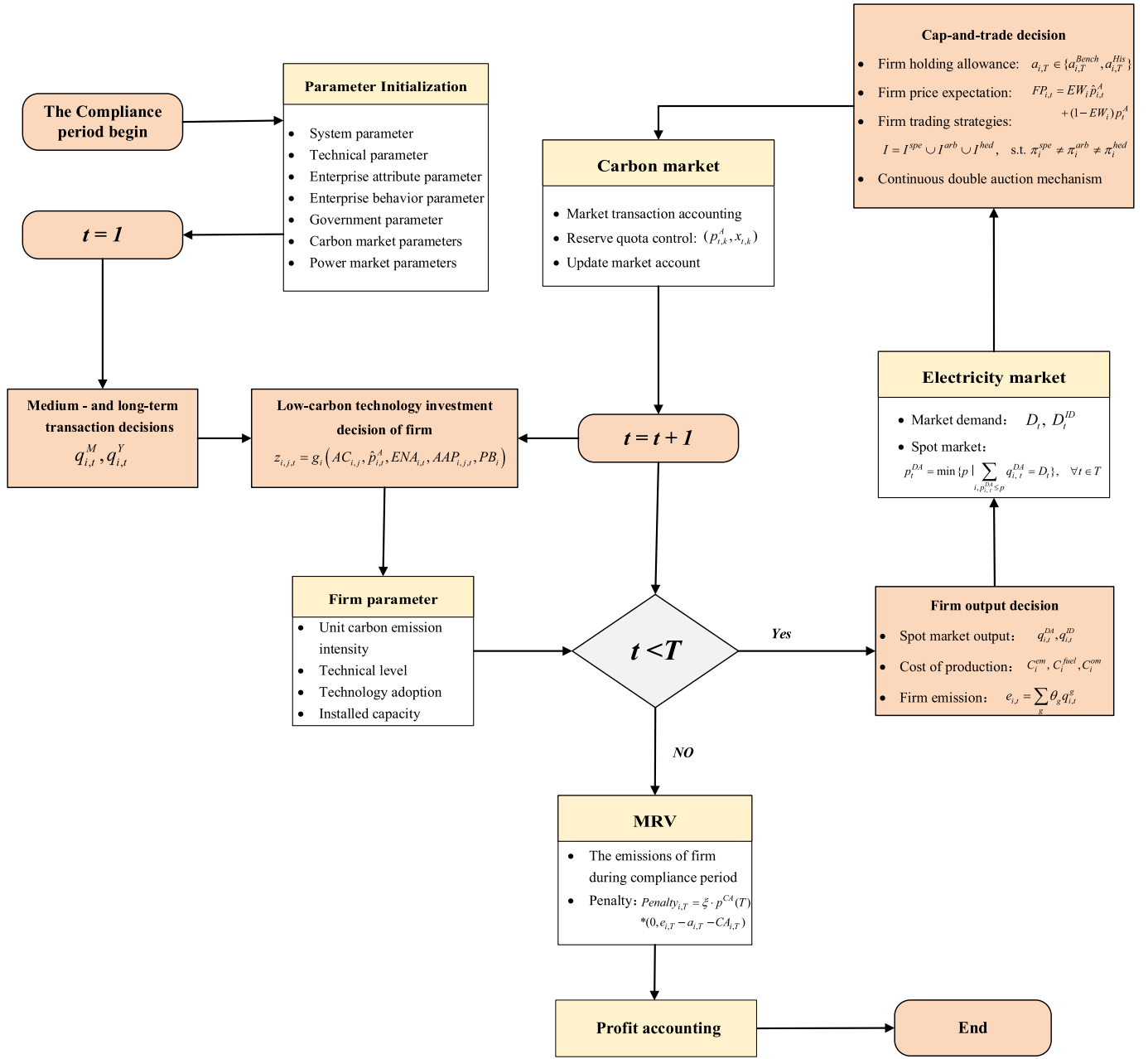


Fig. 1. The flowchart of the model.

$$\left\{ \begin{aligned} C_{i,t} &= C_{i,t}^{em} + C_{i,t}^{fuel} + C_{i,t}^{com} + C_{i,t}^{Act} \\ C_{i,t}^{com} &= \mu_i \cdot IC_i \\ C_{i,t}^{fuel} &= \sum_f q_{i,t}^f \cdot p^f \\ C_{i,t}^{em} &= \sum_k P_{t,k}^A \cdot x_{i,t,k} \\ C_{i,t}^{Act} &= \sum_j CI_j \cdot \frac{T-t}{PB_i \cdot T} \end{aligned} \right.$$

Meanwhile, the firm's low-carbon technology adoption decision, denoted as $z_{i,j,t}$, is contingent upon several factors, including average emission reduction cost ($AC_{i,j}$), the absolute emission reduction potential of technology, the expected allowance price ($FP_{i,t}$), the expected net quota ($ENA_{i,t}$), absolute abatement potential ($AAP_{i,j,t}$) and the firm's expected payback period (PB_i), as detailed in Eq. (4):

$$z_{i,j,t} = g_i(AC_{i,j}, FP_{i,t}, ENA_{i,t}, AAP_{i,j,t}, PB_i) \quad (4)$$

Low-carbon technologies in the model are modeled from a bottom-up perspective, with each technology j characterized by five coefficients: energy-saving cost (ECS_j), investment cost (CI_j), operation cost (MC_j), reduction in emissions (EA_j) and technology lifecycle (TLC_j). ER_i is the expected return of firm i . And the emission abatement cost of each technology ($AC_{i,j}$) can be calculated by Eq. (5):

$$AC_{i,j} = \frac{\left(CI_j + \sum_{y=1}^{PB_i} \frac{MC_j - ECS_j}{(1+ER_i)^y} \right)}{\sum_{y=1}^{TLC_j} EA_j} \quad (5)$$

Building upon this, the model provides equations for calculating the absolute abatement potential of technology j ($AAP_{i,j,t}$), the expected net allowance ($ENA_{i,t}$), and the expected output of firm i in the remaining

period ($EOL_{i,t}$), as shown in Eq. (6)–(8):

$$AAP_{i,j,t} = EOL_{i,t} \cdot EA_j \quad (6)$$

$$ENA_{i,t} = a_{i,t} - EAC_{i,t} - EOL_{i,t} \cdot EI_{i,t} \quad (7)$$

$$EOL_{i,t} = f_i(q_{i,y}^Y, q_{i,m}^M, q_{1,t-1}^{ID}, q_{1,t-1}^{DA}) \quad (8)$$

In Eq. (7), $a_{i,t}$ denotes the quota holding of the firm, $EAC_{i,t}$ represents the cumulative emissions of the firm, and $EI_{i,t}$ means the emission intensity per unit output.

Finally, the model introduces a probability $\psi_{i,t}$ to characterize the propensity of firm i to adopt technology j , which is same to Yu et al. (2020). Eq. (9) presents the technology adoption probability function, indicating that once a firm adopts a low-carbon technology, which can be calculated as:

$$\psi_{i,t} = \left(\frac{1}{1 + \beta_1 \frac{AC_{ij}}{P_{i,t}}} \right)^{\beta_2} \left(\frac{1}{1 + \beta_3 ENA_{i,t}} \right)^{\beta_4} \left(\frac{1}{1 + \beta_5 (-AAP_{i,j,t})} \right)^{\beta_6} \quad (9)$$

2.2. Electricity market module

This paper considers the decision-making behavior of electricity firms in multi-level electricity markets. The spot electricity market can be divided into day-ahead and intraday sub-markets, both of which employ a marginal pricing mechanism. By aggregating the bidding functions of power generation firms, the uniform clearing price and the allocated generation quantity for each firm are determined (Heredia et al., 2018; Zhang and Zhang, 2020; Fang and Ma, 2020; Guo et al., 2024).

In the day-ahead market, the equilibrium clearing price (p_t^{DA}) is determined based on the aggregated bids of electricity firms and market demand (D_t), following the marginal pricing rule:

$$p_t^{DA} = \min \left\{ p \sum_{i: p_{i,t}^{DA} \leq p} q_{i,t}^{DA} = D_t \right\}, \quad \forall t \in T \quad (10)$$

As shown in Eq. (11), the intraday market primarily addresses real-time load fluctuations, allowing firms to adjust the output of generating units based on actual electricity demand and the bids of power generation firms to achieve supply-demand balance in the power system. The equilibrium clearing price of the intraday market (p_t^{ID}) is computed as follows:

$$p_t^{ID} = p_t^{DA} + \alpha \left(D_t^{ID} - \sum_i q_{i,t}^{DA} \right), \quad \forall t \in T \quad (11)$$

Where, D_t^{ID} represents the actual power demand on the day, $\sum_i q_{i,t}^{DA}$ represents the total power generation of the day-ahead spot market, and α is the adjustment coefficient dynamically adjusted by the grid operator.

The medium- to long-term market primarily lock in future electricity prices and quantities through bilateral contracts with multiple time scales, providing both stable expectations for power generation firms and a crucial basis for grid dispatch (Papageorgiou, 2023). Specifically, the monthly contract price (p_m^M), is typically based on the monthly average price of the previous month's day-ahead market ($\bar{p}_{m-1}^{DA}$) and the forecasted electricity demand for the next month ($\hat{D}_m$), reflecting the supply-demand balance in the previous month, as shown in Eq. (12):

$$p_m^M = f \left(\bar{p}_{m-1}^{DA}, \hat{D}_m, \frac{\sum_i q_{i,m}^M}{\hat{D}_m} \right), \quad \forall m \in M \quad (12)$$

The annual contract price (p_y^Y) builds upon the monthly contract

price by incorporating additional factors such as macroeconomic growth rate (GDP_y) and annual electricity demand forecast ($\hat{D}_y$), reflecting the overall economic growth trend and the operational planning of grid operators, as shown in Eq. (13):

$$p_y^Y = g(\bar{p}_y^M, GDP_y, \hat{D}_y), \quad \forall y \in Y \quad (13)$$

2.3. Government module

Based on the current construction of the carbon market and the existing literature, we introduce a multi-level regulatory mechanism to model the government module (Tang et al., 2017; Zhu et al., 2018). At the beginning of the compliance period, the government sets a phased total carbon emission budget (CB_T), which determines the total amount of tradable carbon emissions (A_T) in the carbon market and influences the formation of carbon trading prices.

$$A_T \propto CB_T \quad (14)$$

In current carbon market trading practices, most markets adopt a hybrid allocation method combining free allocation and auctioning (Wang and Zhou, 2020). This study employs the method of Grandfathering as the free allowance allocation method for each firm, and the allocation rules are as follows:

$$q_{i,T}^{Gra} \propto \frac{e_{i,T-1}}{\sum_{i=1}^n e_{i,T-1}}, \quad \forall i \in I \quad (15)$$

The emissions of firms during their compliance period can be calculated according to the emission factor of the power generation technology (θ_i) and the power generation of firm ($q_{i,t}$), as depicted in Eq. (16):

$$e_{i,T} = \sum_t e_{i,t} = q_{i,t}, \quad \forall t \in T \quad (16)$$

Then, based on the firm's emission reductions and actual abatement costs during the compliance period, the government imposes the penalty ($Penalty_{i,T}$) on any excess emissions.

$$Penalty_{i,T} = \xi \cdot p^{CA}(T) \cdot [(0, e_{i,T} - a_{i,T} - CA_{i,T})] \quad (17)$$

$$CA_{i,T} = \sum_t \sum_k x_{i,t,k}, \quad \forall t \in T, \forall k \in K \quad (18)$$

Eq. (18) represents the firm's actual emissions during the compliance period, $CA_{i,T}$ is the total amount of allowances traded by the firm in the carbon market, $p^{CA}(T)$ is the average carbon price during the compliance period, and ξ is the penalty coefficient.

2.4. Carbon market module

The carbon market module is a crucial element of the ABM model for depicting the operational mechanism of the ETS (Richstein et al., 2014; Tang et al., 2015). A continuous double auction mechanism is employed to model the trading process, determining the trading strategies of both buyers and sellers. Buyers and sellers determine their bidding/asking prices ($p_{i,t,k}^{A,bid/ask}$) and volumes ($x_{i,t,k}$) based on their expected carbon price, as depicted in Fig. 2.

Based on the continuous double auction mechanism, the market matches the submitted orders with existing orders, continuously executing trades according to the price-time priority rule, and forming the actual transaction price and quantity at that moment (Zhang et al., 2011). Moreover, the market fully considers the heterogeneous behavioral preferences of different trading entities, demonstrating strong inclusivity (Yu et al., 2020).

To refine the formation mechanism of expected carbon prices, the model sets the technical expected carbon price ($\hat{p}_{i,t}^A$) based on factors

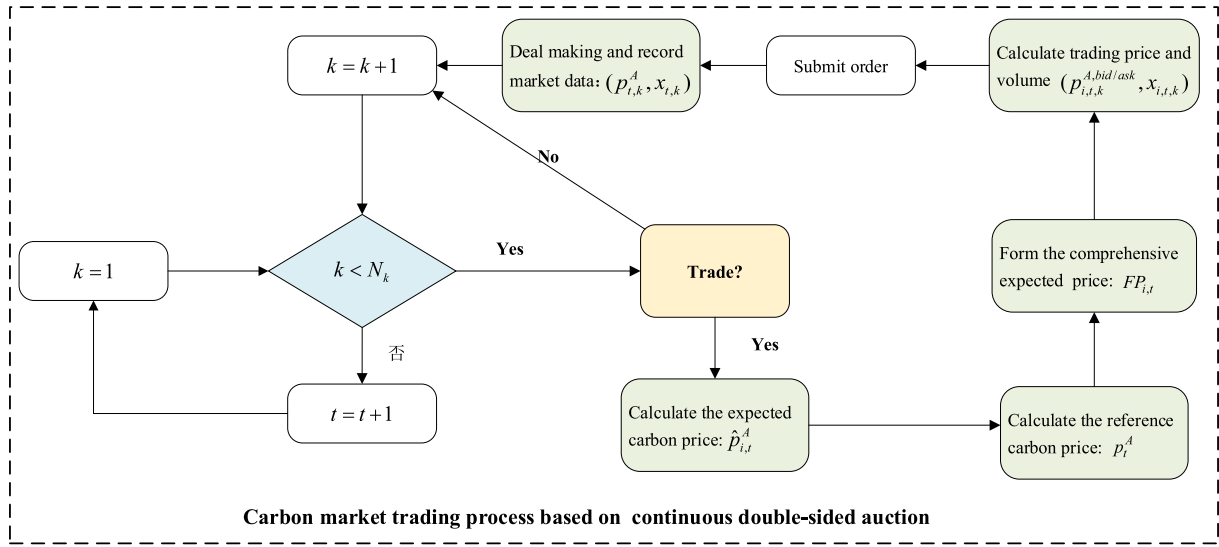


Fig. 2. Time frame of the allowance trading process.

such as the firm's emissions intensity per unit of output, expected net allowance, historical utility function, and firm utility function, as shown in Eq. (19).

$$\hat{p}_{i,t}^A = f(El_{i,t}, ENA_{i,t}, \pi_{i,1:t-1}, u_i), \quad \forall i \in I, \forall t \in T \quad (19)$$

The model also defines the market-based expected reference price (p_t^A), which is determined by the historical price trends ($p_{1:t-1}^A$) and market supply and demand expectations, as shown in Eq. (20). Finally, to form a comprehensive expected allowance price ($FP_{i,t}$), firms define it as a weighted average of the expected carbon price ($\hat{p}_{i,t}^A$) and the reference price (p_t^A), as shown in Eq. (21):

$$p_t^A = w_1 p_{1:t-1}^A + w_2 \frac{\sum_i d_{i,t}^A}{A_t} \quad (20)$$

$$FP_{i,t} = EW_i \hat{p}_{i,t}^A + (1 - EW_i) p_t^A \quad (21)$$

where, $d_{i,t}^A = \max\{x_{i,t}, 0\}$ reflects the net quota demand of the firm i in the period t . A_t is the total quota for period t , and the supply-demand ratio $\frac{\sum_i d_{i,t}^A}{A_t}$ reflects the market sentiment. w_1 and w_2 are the inertia coefficient and the supply and demand coefficient.

According to the strategy function, the declaration of trading firms constitutes the market's bid price ($BID_{t,k}$) and ask price ($ASK_{t,k}$), as shown in Eq. (22) and Eq. (23):

$$BID_{t,k} = (p_{i,t,k}^{A,bid}, x_{i,t,k}) : p_{i,t,k}^{A,bid} = p_{t,k}^A + \Delta p_{i,t,k}^A, \Delta p_{i,t,k}^A < 0 \quad (22)$$

$$ASK_{t,k} = (p_{i,t,k}^{A,ask}, x_{i,t,k}) : p_{i,t,k}^{A,ask} = p_{t,k}^A + \Delta p_{i,t,k}^A, \Delta p_{i,t,k}^A > 0 \quad (23)$$

The conditional probability of the declared price spreads ($\Delta p_{i,t,k}^A$) is the percentage of the declared price above or below the reference price. And it is a crucial parameter reflecting the heterogeneous trading strategies adopted by firms. This parameter determines the percentage by which a firm's declared bid or ask price deviates above or below a dynamically determined reference price, influenced by factors such as their policy tendency to hold quotas and their speculative tendencies.

$$\Delta p_{i,t,k}^A = \begin{cases} 10, & \text{if } p_t^A \leq \hat{p}_{i,t}^A - \beta_i a_{i,t,k} - \gamma_i (p_t^A - p_{t-1}^A) \\ 8, & \text{if } \hat{p}_{i,t}^A - \beta_i a_{i,t,k} - \gamma_i (p_t^A - p_{t-1}^A) < p_t^A \leq \hat{p}_{i,t}^A \\ -8, & \text{if } \hat{p}_{i,t}^A < p_t^A \leq \hat{p}_{i,t}^A + \beta_i a_{i,t,k} + \gamma_i (p_t^A - p_{t-1}^A) \\ -10, & \text{if } p_t^A > \hat{p}_{i,t}^A + \beta_i a_{i,t,k} + \gamma_i (p_t^A - p_{t-1}^A) \end{cases} \quad (24)$$

In Eq. (24), β_i denotes the policy tendency of firms to hold quotas (i.e. wait-and-see strategy), influencing whether their declared price will be higher (if selling) or lower (if buying) to reflect a less urgent trading posture. And $a_{i,t,k}$ represents the number of quotas held by the firm. Conversely, γ_i represents firm's tendency to employ speculative strategies, leading to larger declared price spreads aimed at maximizing short-term gains or exploiting perceived market mispricing. The specific discrete thresholds (e.g., $\pm 8\%$, $\pm 10\%$) for these price spreads were derived from an empirical analysis of historical price fluctuations and observed trading behavior within the Chinese carbon market data. These values represent typical price deviations that trigger distinct strategic responses among market participants, differentiating aggressive trading from more conservative approaches.

These declared price spreads are then fed into a continuous double-sided auction mechanism, which forms the core of our secondary market for carbon emission allowances. The mechanism integrates the price orders in the secondary market for carbon emission allowances using the trading order book mechanism introduced by Yu et al. (2020). And the orders are arranged according to the principle of price priority and time priority.

The decision-making strategies of different firms exhibit significant heterogeneity. To fully capture this characteristic, our model characterizes heterogeneity from the perspective of trading preferences. Based on differences in trading motives, traders can be categorized into three types: speculative, arbitrage, and risk-hedge, as shown in Eq. (25). Their utility functions and decision criteria differ substantially.

$$I = I^{spe} \cup I^{arb} \cup I^{hed}, \quad \text{s.t. } \pi_i^{spe} \neq \pi_i^{arb} \neq \pi_i^{hed}, \quad \forall i \in I \quad (25)$$

Moreover, the utility functions of different types of traders are distinct. For example, speculative traders seek to maximize price spread profits, and their utility function can be expressed as Eq. (26):

$$u_i^{spe}(\Delta p_{i,t,k}^A, x_{i,t,k}) = E[(p_t^A - p_{t,k}^A) x_{i,t,k}] - \lambda Var[(p_t^A - p_{t,k}^A) x_{i,t,k}] \quad (26)$$

The decision of a speculator is essentially a risk-return optimization problem, and Eq. (26) states that the utility of trader i is determined by

the mean ($E[(p_T^A - p_{t,k}^A)x_{i,t,k}]$) and variance ($Var[(p_T^A - p_{t,k}^A)x_{i,t,k}]$) of his expected spread return. p_T^A represents the expected carbon price at maturity and λ is the coefficient of risk aversion.

In contrast, arbitrage traders earn profit from price discrepancies in the market, while risk-hedge traders seek to minimize their risk exposure.

$$u_i^{arb}(\Delta p_{i,t,k}^A, x_{j,t,k}) = E[(p_{t_2}^A - p_{t_1}^A)x_{i,t,k}] - \lambda Var[(p_{t_2}^A - p_{t_1}^A)x_{i,t,k}] \quad (27)$$

$$u_i^{hed}(\Delta p_{i,t,k}^A, x_{i,t,k}) = -\lambda Var[(p_T^A - p_{t,k}^A)x_{i,t,k}] \quad (28)$$

Based on game theory, the model solves the equilibrium carbon price under the constraint of maximizing the utility of heterogeneous traders. Eq. (29) indicates that the equilibrium price in the carbon market is the price level that maximizes the utility of all traders and equates the total bid quantity of buyers with the total bid quantity of sellers.

$$\begin{aligned} \max_{\Delta p_{i,t,k}^A} & \Delta p_{i,t,k}^A(x_{i,t,k}) \\ \text{s.t.} & \sum_{i \in I^{bid}} x_{i,t,k} = \sum_{j \in I^{ask}} x_{j,t,k}, \end{aligned} \quad (29)$$

$$p_t^A = p_{t,k}^A : \sum_{i \in I^{bid}} x_{i,t,k}(p_t^A) = \sum_{j \in I^{ask}} x_{j,t,k}(p_t^A)$$

2.5. Optimization of multi-agent reinforcement learning algorithm

To comprehensively characterize the dynamic evolution of the electricity-carbon coupled market, this study introduces targeted improvements and innovative applications to the MADDPG algorithm within the MARL framework. Specifically, by incorporating the adaptive strategy such as Proximal Policy Optimization (PPO) and attention mechanism, we design an intelligent game optimization algorithm that balances heterogeneity, cooperation, and generalization to solve for the equilibrium evolution path and key influencing factors of the electricity-carbon system (Harrold et al., 2022; Zhou et al., 2024).

Mathematically, we can reformulate the underlying optimal control problem as a formal MARL problem using Markov games, which involves a set of agents, each with its own state space, action space, reward set, and transition probability distribution (Qiu et al., 2023; Zhou et al., 2024). The model first defines the state space, local observation, and action space. In the learning process based on the Actor-Critic framework, the policy network refines its output by learning and adjusting the parameters of the actor network (θ_i^μ), enabling it to select the optimal action in a given state to maximize long-term rewards.

$$act_{i,t} \sim \mathcal{N}(\mu_i(o_{i,t}|\theta_i^\mu), \sigma_i^2) \quad (30)$$

Eq. (30) states that in a state $o_{i,t}$, the policy network μ_i gives a deterministic action $\mu_i(o_{i,t}|\theta_i^\mu)$. Then the final random action $act_{i,t}$ is obtained through Gaussian noise disturbance $\mathcal{N}$ with variance of σ_i^2 . In addition, the intelligent algorithm is optimized from the following aspects.

Considering the continuous and high-dimensional action space of the agents, as well as the balance between exploration and exploitation, we introduce the PPO algorithm to optimize the policy gradient of the actor network based on the traditional DDPG architecture. With the PPO algorithm, we construct the following policy objective function ($J_i^\pi(\theta)$):

$$J_i^\pi(\theta) = \hat{E}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)] \quad (31)$$

In Eq. (31), $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio, $\hat{A}_t$ denotes the generalized dominant function, and ϵ is a hyper-parameter. $r_t(\theta)\hat{A}_t$ represents the advantage of the current policy over the old policy, while the opposite limits the update magnitude of the policy. It is an advantage function corrected by importance sampling, which encourages exploration that is better than the old policy while avoiding excessive policy

deviation. Furthermore, we use the stochastic gradient ascent algorithm to optimize the objective function, thereby updating and iterating the parameters of the policy network. And the workflow of PPO-clip is summarized in Algorithm 1.

$$\begin{aligned} \nabla_{\theta} J_i^\pi(\theta) & \approx \frac{1}{T} \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_{i,t}|s_t) \min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \\ \theta_i' & \leftarrow \theta_i + \alpha \nabla_{\theta} J_i^\pi(\theta_i') \end{aligned} \quad (32)$$

Algorithm 1: PPO-Clip

- 1: Input: Initializes the policy parameter and the value function parameter
- 2: For $i = 0, 1, 2, \dots$ do
- 3: Collect the trajectory set of value functions, by running policies in the environment
- 4: Calculate the probability ratio: $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$
- 5: Estimate the advantage function (using time series difference method): $\hat{A}_t$
- 6: Maximize the PPO-clip target and calculate the gradient of the objective function $\nabla_{\theta} J_i^\pi(\theta)$: $\nabla_{\theta} J_i^\pi(\theta) \approx \frac{1}{T} \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_{i,t}|s_t) \min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)$
- 7: Update the strategy using the gradient ascent method: $\theta_i' \leftarrow \theta_i + \alpha \nabla_{\theta} J_i^\pi(\theta_i')$
- 8: Update the value function
- 9: end for

Eq. (32) represents the updating process of the policy network parameters of agent i . Firstly, based on the sampled trajectory data, the gradient of the objective function to the parameters ($\nabla_{\theta} J_i^\pi(\theta)$) is calculated according to the strategy gradient theorem, and then the new parameters (θ_i') are obtained by gradient rise with the learning rate (α).

To model the correlation weights between agents and enhance the information extraction and value function fitting capabilities of the Critic network, we introduce an attention mechanism into the model, as shown in Eq. (33):

$$\alpha_{i,t} = \frac{\exp(\text{score}(f_s(s_t), h_i(a_{i,t})))}{\sum_{j=1}^N \exp(\text{score}(f_s(s_t), h_j(a_{j,t})))} \quad (33)$$

where, f_s and h_i are encoding functions for state features and action features, respectively, and $\text{score}(\cdot)$ is the similarity scoring function between the two feature vectors. $\alpha_{i,t}$ can be interpreted as the degree of attention paid to the agent i at time t based on global information, which is used to weight the importance of different agents in value function estimation. And the update of the value function (θ_i^Q) involves the following three stages:

$$\begin{aligned} \tilde{Q}(s_t, a_{1,t}, \dots, a_{N,t}) & = \sum_{i=1}^N \alpha_{i,t} Q_i(s_t, a_{1,t}, \dots, a_{N,t}) \\ \mathcal{L}(\theta_i^Q) & = \mathbb{E}_{s,a,r,s'}[r] \\ \theta_i^Q & \leftarrow \theta_i^Q - \beta \nabla_{\theta_i^Q} \mathcal{L}(\theta_i^Q) \end{aligned} \quad (34)$$

First, the value function estimates of each agent are weighted based on the attention weights to obtain a joint value function $\tilde{Q}$; Then, $\mathcal{L}(\theta_i^Q)$ is calculated based on temporal difference learning, where $\tilde{Q}(s, a_{1,t}, \dots, a_{N,t}; \theta_i^Q)$ is the current value estimate; Finally, the gradient descent method is used to update the value function θ_i^Q based on the loss function.

Overall, based on the MADDPG algorithm, the optimization part focuses on “strategy learning of heterogeneous agents, cooperative game, and joint optimization”, and proposes a “PPO-Attention” fusion enhancement paradigm. Following multiple iterations and training

sessions, the strategy of agents progressively converges towards a stable state, facilitating enhanced collaborative learning and interaction among agents. And the constructed intelligent optimization algorithm can significantly improve the modeling and solving capabilities of multi-agent game in the electricity-carbon system.

2.6. Data resource and training results

The simulation program is executed on the NVIDIA GeForce RTX 4090 and implemented within the Python 3.12 framework. Based on real-world data from the *Global Power Enterprise Database* and China's Power Statistical Yearbook. This study simulated the decision-making processes and behavior of 878 million-kilowatt coal-fired power generation firms in China, setting heterogeneous parameters for each enterprise in terms of scale, cost, technological level, and regional resource endowment. Besides, this paper trains the agents using the real price data from the China national carbon market, the electricity market, and energy markets.

Fig. 3 illustrates the comparative performance of three reinforcement learning algorithms—PPO-MADDPG, MADDPG, and DDPG—in the electricity-carbon system multi-agent game scenario. The results indicate that the PPO-MADDPG algorithm significantly outperforms both MADDPG and DDPG in terms of final converged average reward value and convergence speed. This superior performance stems from PPO-MADDPG's integration of PPO clipping mechanisms, which stabilize policy updates by constraining gradient deviations, and attention mechanisms that enhance coordination among agents by dynamically weighting their interactions. In contrast, DDPG exhibits the lowest reward values and unstable convergence due to its reliance on independent learning, which fails to account for the interdependencies between agents in dynamic multi-agent environments.

During the initial exploration phase, all algorithms show fluctuating rewards as agents iteratively adapt their strategies. However, DDPG's oscillations persist throughout training (reward variance: ± 0.35 vs. ± 0.18 for PPO-MADDPG), reflecting its inability to disentangle the compounded effects of concurrent policy adjustments across agents. While MADDPG and PPO-MADDPG also experience fluctuations (e.g., episodic reward drops of $\sim 15\%$ at mid-training), these are inherent to multi-agent systems where each agent's reward depends on the evolving strategies of all others. Specifically, as agents update their policies, the perceived environment dynamics shift, creating temporary instability until equilibrium strategies emerge.

The dominance of PPO-MADDPG underscores its suitability for high-dimensional, cooperative-competitive environments like electricity-carbon markets. Its reward advantage aligns with prior studies that highlight the importance of constrained policy optimization in multi-agent settings. These findings validate the algorithmic enhancements proposed in this work and emphasize the necessity of coordinated learning frameworks to address the complexity of heterogeneous firm interactions in carbon markets.

3. Results and discussions

3.1. The baseline results of trading and adoption of low-carbon technologies

Fig. 4 illustrates the simulated short-term price and trading volume dynamics in the carbon market, revealing distinct phases of fluctuation. Initially, the carbon price surged rapidly from its starting point to a peak of approximately 120 CNY/ton, driven by heightened demand from firms adjusting their production forecasts and allowance-holding strategies. During the second phase, reduced trading activity led to a sharp decline and subsequent stabilization of the allowance price at around 50 CNY/ton, reflecting a temporary weakening in market demand. As the compliance deadline approached, firms engaged in intensive allowance purchases to meet regulatory requirements, driving the price to a peak of approximately 190 CNY/ton. These pronounced price fluctuations underscore the market's "compliance-driven trading" behavior, characterized by concentrated trading activities near compliance deadlines, resulting in significant volatility.

The adoption of low-carbon technologies by firms demonstrates significant temporal and scale-related disparities. During the compliance period, as depicted in Fig. 5, firms adopted a total of 425 low-carbon technologies, with 80 % of these adoptions concentrated within the initial 20 % of the compliance timeline. This early adoption trend aligns with Yu et al. (2020), who highlighted that diminishing returns from emission reductions and escalating marginal costs deter continuous adoption over time. Furthermore, Fig. 5 reveals that large-scale power generation firms, categorized by higher installed capacities, dominate technology adoption efforts, while small and medium-scale firms delay adoption. This divergence stems from resource disparities: large-scale firms leverage robust financial capacity, advanced technological reserves, and market influence to lead in low-carbon innovation. In contrast, small and medium-scale firms face capital and

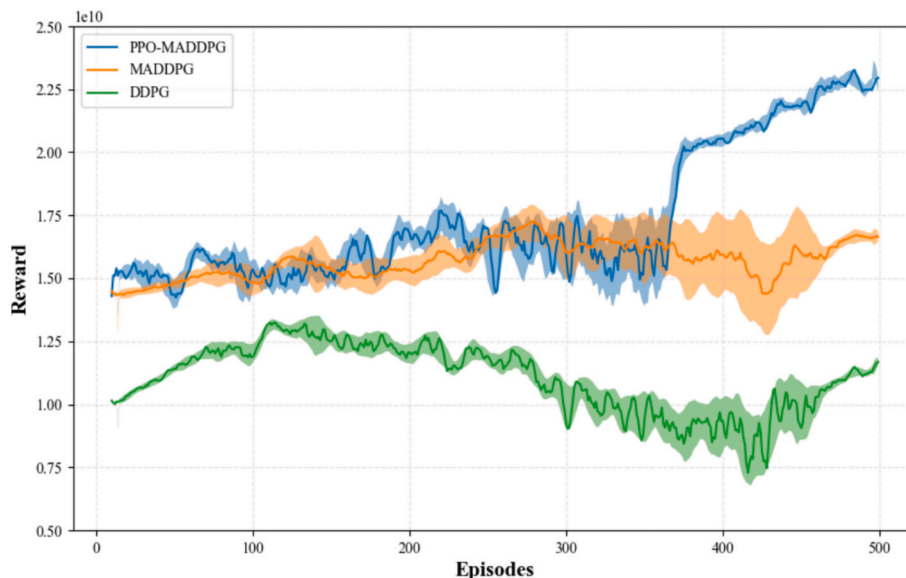


Fig. 3. The convergence curve of reward function under different algorithms.

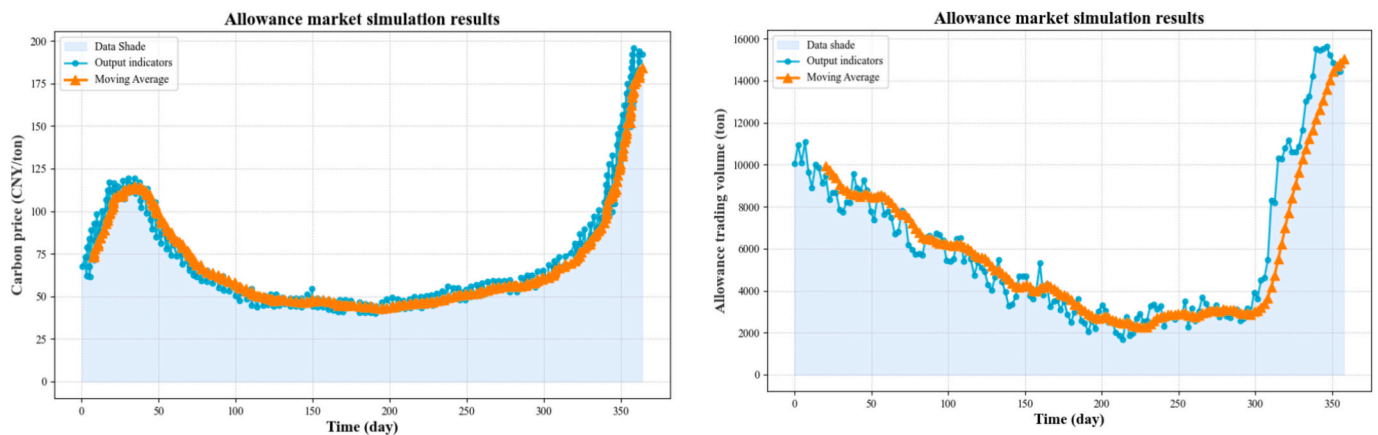


Fig. 4. The dynamics of allowance prices (left) and allowance trading volume (right).

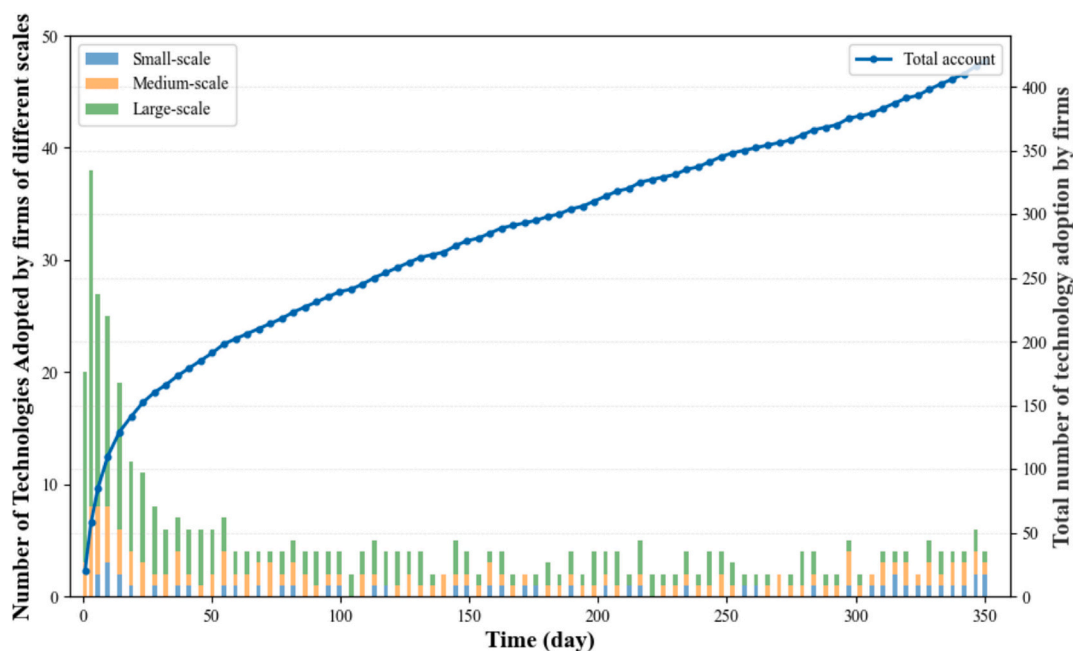


Fig. 5. The adoption of low-carbon technologies for different scales of firms. Note. The model incorporates 150 low-carbon technologies for thermal power generation, with the initial adoption levels across firms distributed according to a Pareto distribution.

technological constraints, resulting in a more cautious approach to adopting low-carbon solutions. In addition, we conducted a supplementary analysis of the heterogeneity of low-carbon technologies. The relevant results are detailed in Appendix B.

3.2. The influence of the expected payback period

Fig. 6 illustrates the influence of the expected payback period for low-carbon investments on the average carbon price and allowance trading volume in the carbon market. As the expected payback period increases, the average carbon price demonstrates a clear downward trend, decreasing from approximately 170 CNY/ton to 90 CNY/ton. This decline suggests that longer payback periods fundamentally alter firms' investment calculus and compliance strategies. When low-carbon investments (e.g., energy efficiency upgrades, carbon capture technologies) promise returns over a more extended timeframe, the immediate financial incentive for firms to reduce their carbon emissions through active abatement or costly allowance purchases diminishes. This is because a longer payback period implies a lower net present value for a given investment in the short-to-medium term, reducing the perceived

urgency of reducing emissions through technology adoption. Consequently, firms with longer investment horizons are less compelled to acquire allowances to cover emissions, reducing overall demand in the carbon market and pushing prices downwards. While the average allowance trading volume remains relatively stable, fluctuating around 480–510 million tons, indicating its compliance-driven nature, the price dynamics are highly sensitive to these economic expectations. This contrast highlights that even if firms trade to fulfill regulatory obligations, the intensity of demand (and thus price) is heavily influenced by the perceived economic burden and return on emission reduction efforts.

Fig. 7 further illustrates the nuanced influence of the expected payback period on the adoption of low-carbon technologies across firms of different scales. Notably, the total number of technology adoptions increases as the expected payback period lengthens, with large-scale firms driving the majority of adoptions. This seemingly counter-intuitive observation is underpinned by key economic mechanisms. Longer payback periods, while dampening immediate carbon price signals, create an environment where the long-term strategic value of innovative low-carbon technologies becomes more prominent.

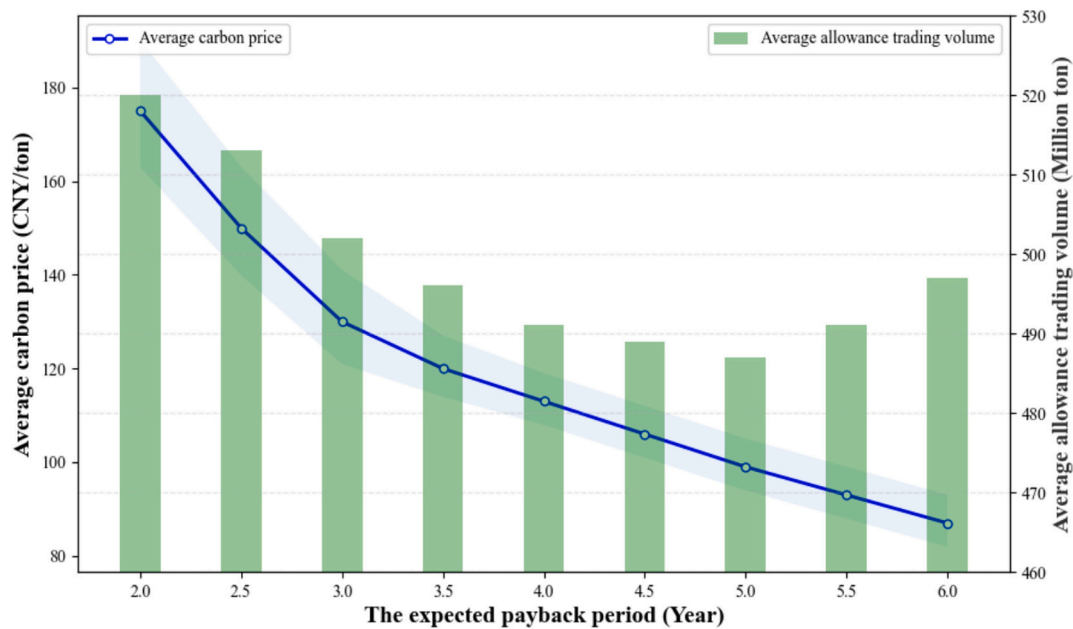


Fig. 6. The influence of the expected payback period on allowance trading.

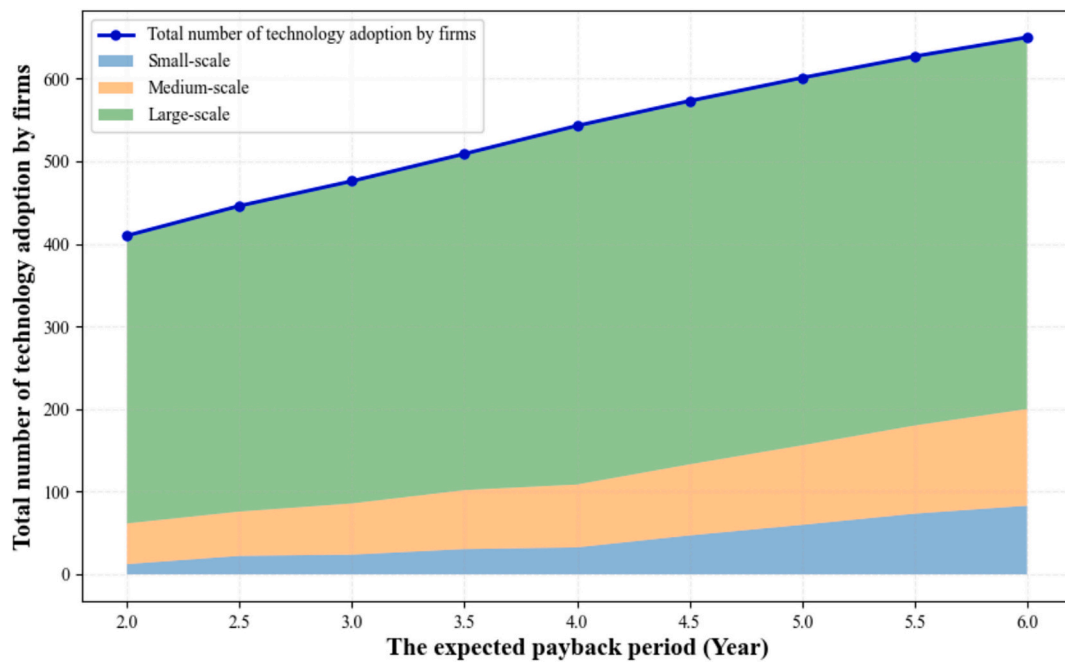


Fig. 7. The influence of the expected payback period on the adoption of low-carbon technologies for different scales firms.

Innovative technologies (easier for large firms, as discussed in Appendix B) typically involve higher upfront investment costs and longer lead times for returns, but also offer deeper and more sustainable emission reductions. For large firms, with their robust financial resources, lower capital constraints, and longer strategic planning horizons, the extended payback period makes these deeper, more transformative investments economically justifiable. They are better positioned to absorb the initial high costs and wait for the long-term benefits, which include not only compliance but also competitive advantage and reduced future operational costs. In contrast, medium-scale firms show a moderate increase in adoption with the extension of the payback period, reflecting their relatively limited but steady capacity for low-carbon investments, often preferring supplementary technologies with clearer and more

predictable returns. Small-scale firms demonstrate the lowest adoption levels, with a minimal contribution to the total adoption. This highlights their acute financial and technological constraints; for them, a longer payback period exacerbates risk and cash flow issues, making any significant low-carbon investment prohibitive unless directly incentivized.

These results emphasize that resource endowments and firm scale significantly influence the pace and extent of low-carbon technology adoption. Policies aimed at accelerating sustainable technology adoption should therefore consider tailored support mechanisms. While a higher carbon price can incentivize short-term abatement, policies that reduce the perceived financial risk and provide targeted financial incentives (e.g., subsidies, low-interest loans, or favorable tax treatments) are crucial for mitigating the barriers posed by extended payback

periods. Such measures are particularly vital for enabling broader participation across all firm scales, especially for small and medium-sized firms, and for accelerating the adoption of truly transformative, innovative low-carbon technologies essential for deep decarbonization.

3.3. The influence of the technical expectation weight

Fig. 8 illustrates the influence of the technical expectation weight of firms on the average carbon price and allowance trading volume. As the technical expectation weight increases, both the average carbon price and the average trading volume show an upward trend. The carbon price starts at approximately 70 CNY/ton when the technical expectation weight is low and rises steadily, reaching over 110 CNY/ton at higher technical expectation weights. This increase reflects the market's response to firms' higher expectations for technical improvements, as firms are more willing to invest in carbon trading due to their confidence in achieving emission reductions through technological advancements. Similarly, the allowance trading volume grows progressively, starting below 500 million tons and surpassing 750 million tons as the technical expectation weight increases. The rise in trading volume indicates that firms with higher technical expectations actively engage in allowance trading, potentially leveraging their technological capabilities to optimize compliance strategies. These results highlight the critical role of technical expectations in shaping market behavior. Firms with higher confidence in technological progress are more inclined to participate actively in the carbon market, driving both price and trading volume upward. This underscores the importance of fostering innovation and technological optimism in promoting active market participation and achieving emission reduction targets.

Fig. 9 illustrates the impact of the technical expectation weight of firms on the total adoption of low-carbon technologies, categorized by firm scale. The total number of technology adoptions increases steadily as the technical expectation weight rises, indicating a positive correlation between firms' confidence in technological advancements and their adoption of low-carbon technologies. Initially, at low technical expectation weights (around 0.1), the total adoption level is relatively low, with large-scale firms contributing the most, followed by medium- and small-scale firms. As the technical expectation weight grows, the total adoption expands significantly, particularly driven by large-scale firms. This dominance of large-scale firms suggests that they are better

equipped or more willing to invest in low-carbon technologies when they perceive greater technological certainty. Medium-scale firms also show a moderate increase in adoption, while small-scale firms contribute minimally throughout the range. At higher technical expectation weights (above 0.8), the total adoption plateaus, approaching saturation. This suggests that further increases in technical expectation weight yield diminishing returns in terms of new technology adoptions, possibly due to market saturation or cost constraints. The findings highlight the critical role of technical expectations in promoting the adoption of low-carbon technologies, particularly among large-scale firms.

3.4. The influence of trading preferences

Fig. 10 presents the trading behavior of speculative, arbitrage, and risk-hedge firms in the carbon market, highlighting their differentiated roles and impacts on trading volume, carbon price stability. Arbitrage firms, which capitalize on price discrepancies within the market, demonstrate the highest trading volumes among the three types of firms. Their frequent trading activity, driven by short-term price differentials, leads to pronounced price fluctuations, particularly at the beginning and end of trading periods. This behavior underscores their significant influence on the short-term volatility of carbon prices and market liquidity. Risk-hedge firms, in contrast, prioritize reducing their exposure to price risks. Their trading behavior is conservative and stable, with trading volumes evenly distributed throughout the year. This consistency supports baseline market liquidity and mitigates extreme price volatility, contributing to a more stable and predictable carbon price signal. Such stability is critical for fostering confidence in the carbon market and encouraging long-term investment in emissions reduction. Speculative firms, which aim to profit from long-term price movements, exhibit lower trading frequencies compared to arbitrageurs but engage in higher-risk trading strategies. Their activities have a more pronounced impact on long-term price trends, though their influence on short-term price fluctuations remains relatively limited due to their lower trading frequency.

Fig. 11 highlights the firms' characteristics in low-carbon technology adoption. The daily adoption rates for all three groups show significant variability in the early days, with spikes occurring most prominently for Hedgers. Over time, the fluctuations reduce, stabilizing at relatively

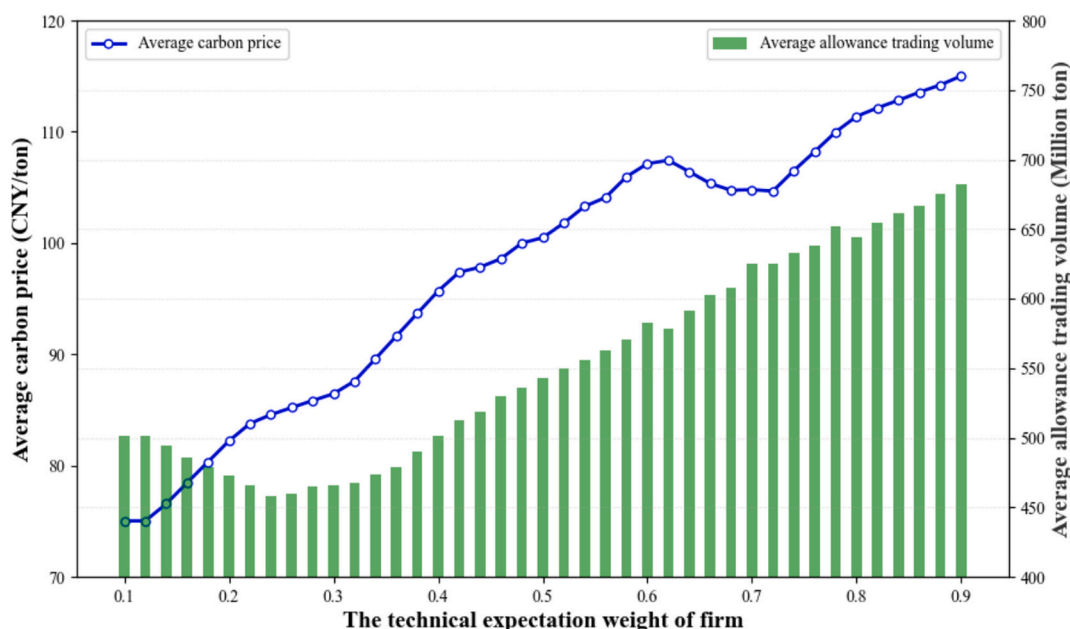


Fig. 8. The influence of the technical expectation weight on allowance trading.

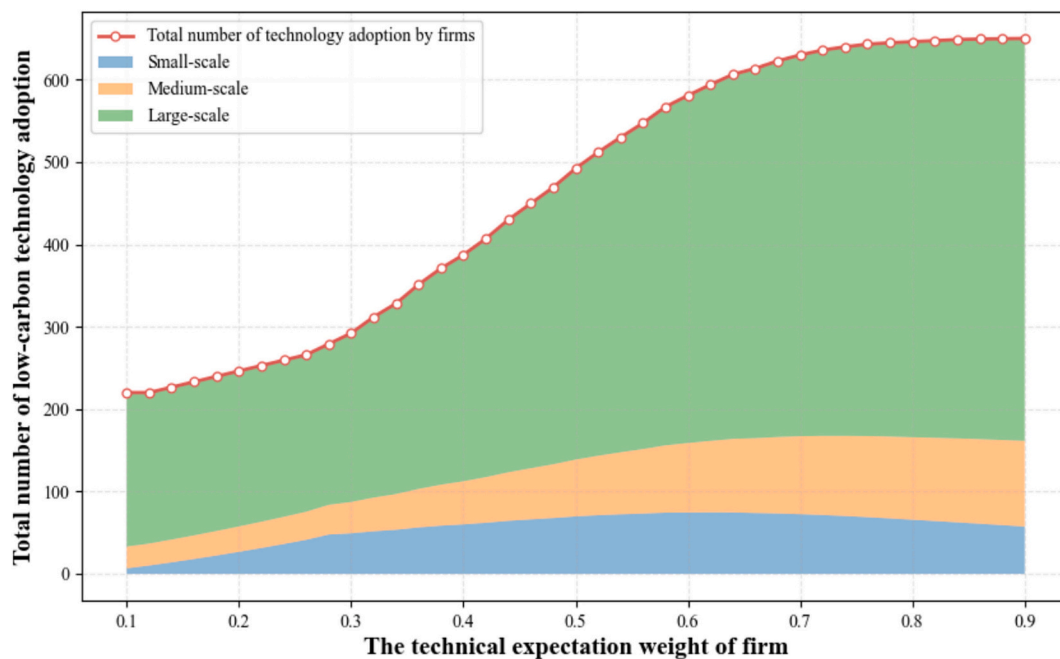


Fig. 9. The influence of the technical expectation weight on the adoption of low-carbon technologies.

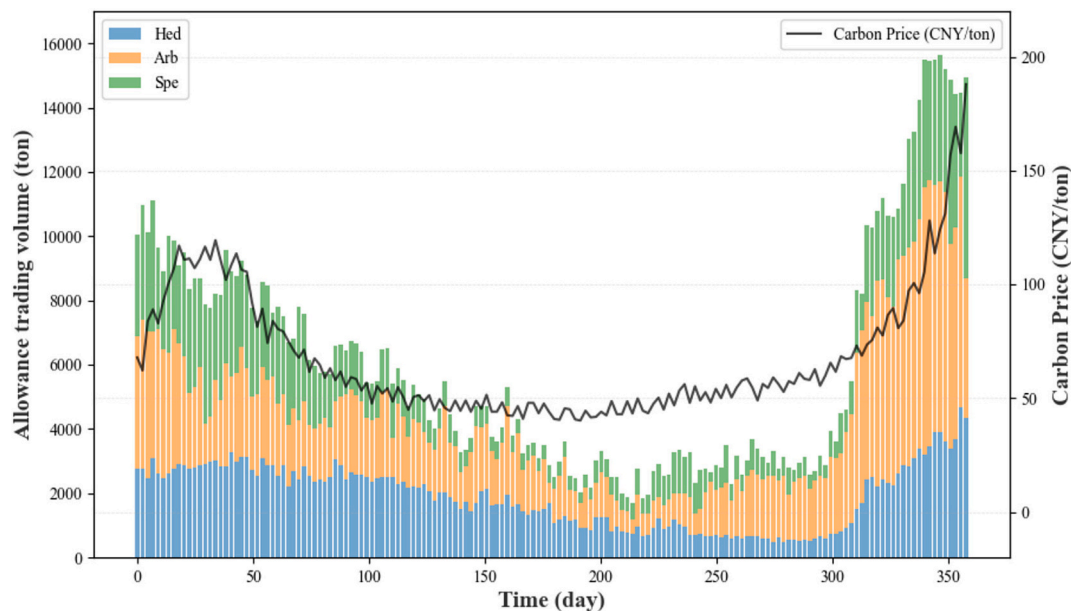


Fig. 10. The influence of the trading preferences on the allowance trading.

lower levels compared to the initial phase. Hedgers exhibit the highest daily adoption rates, particularly during the early phase of the timeline, indicating a preference for rapid adoption. Speculators and Arbitrageurs have comparatively lower and more evenly distributed daily adoption rates throughout the observed period. The cumulative adoption trajectories reveal a consistent ranking among the groups. Hedgers have the highest total adoption by the end of the observation period, followed by Speculators, and finally Arbitrageurs. The figure highlights the critical role of trading preferences in shaping the adoption dynamics of low-carbon technologies. Hedgers, driven by a preference for risk minimization, demonstrate the highest propensity for adopting these technologies. This behavior suggests that risk mitigation strategies are strongly aligned with environmental investments. Conversely, Arbitrageurs, who prioritize profit maximization from price discrepancies, show relatively

low adoption rates, potentially reflecting a weaker alignment between arbitrage opportunities and environmental priorities. Speculators exhibit intermediate behavior, reflecting a mix of opportunistic and strategic engagement with low-carbon technologies.

4. Conclusion

This paper proposes an electricity-carbon coupled market trading mechanism model based on the ABM and MARL methods. Using real-world calibrated data from Chinese power generation firms, the electricity market, the carbon market, and the energy market, we simulate the decisions of heterogeneous electricity firms with different preferences regarding allowance trading and low-carbon technology investment. Our findings reveal that:

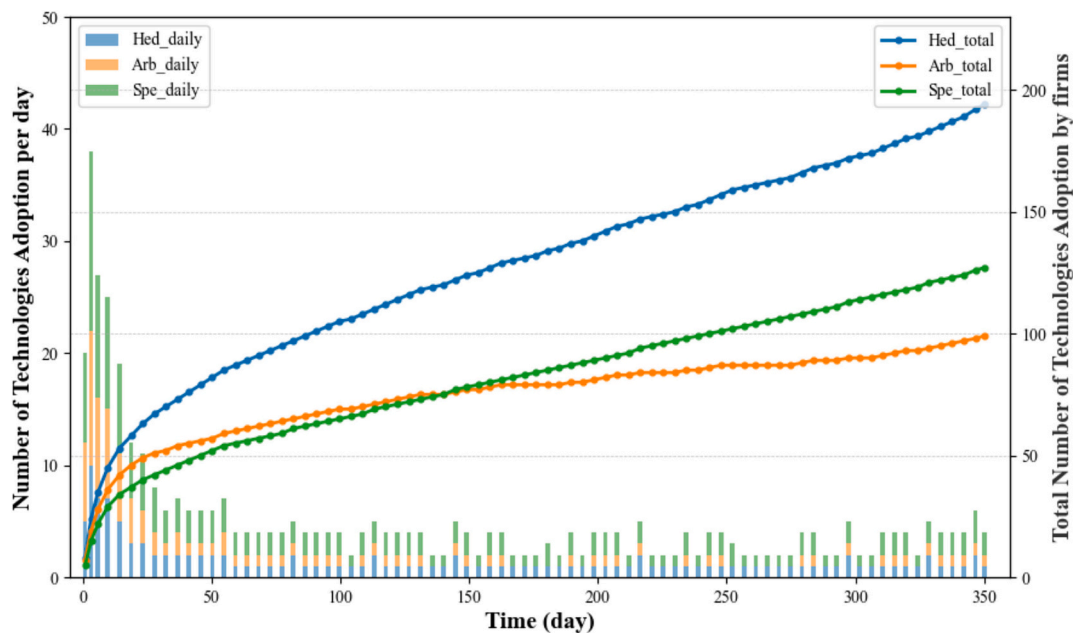


Fig. 11. The influence of the trading preferences on the adoption of low-carbon technologies.

First, there is a significant “compliance-based trading” behavior among regulated entities in the carbon market, where firms tend to conduct large-scale transactions around the compliance period to meet their compliance requirements, leading to a surge in both the volume and price of the carbon market. This is also consistent with the behavior of regulated entities in the China national carbon market. Second, the adoption of low-carbon technologies mainly occurred in the earlier stages of simulation, where large-scale firms have stronger financial strength and technical reserves, and had a significant first-mover advantage in adopting low-carbon technologies. Third, firms’ expected payback period and technical expectation weights have a significant impact on carbon trading and low-carbon technology innovation. A longer expected payback period for firms leads to lower carbon prices and trading volumes but promotes low-carbon technology innovation; a higher technology-oriented expectation weight can encourage regulated entities to participate in allowance trading, stimulate carbon price increases, and promote low-carbon technology innovation. Fourth, trading preferences significantly influence both carbon market dynamics and low-carbon technology adoption, with arbitrage firms driving short-term price volatility but showing low adoption rates, risk-hedge firms contributing to market stability and leading in technology adoption, and speculative firms balancing long-term price impacts with moderate technology engagement.

Our recommendations include: (1) To mitigate “compliance-based trading” behavior, it is essential to enhance transparency within the carbon market by providing more accurate and timely information about compliance requirements and market forecasts. Policymakers should consider improving the communication of carbon allowance allocations, future market conditions, and regulatory changes well in advance of the compliance period. Additionally, strengthening regulatory oversight to monitor trading activities, especially during peak trading periods, can help prevent excessive volatility driven by compliance behavior and speculative trading. (2) To encourage broader adoption of low-carbon technologies, the government should provide

targeted financial incentives such as subsidies, tax credits, or low-interest loans to smaller and medium-sized firms. Moreover, a dedicated technology transfer program could assist these firms by offering technical assistance, training, and access to cutting-edge low-carbon solutions, thus leveling the playing field and accelerating the transition to cleaner energy. (3) Policymakers should consider refining carbon market design to account for these heterogeneous behavior. For instance, introducing measures such as differentiated auctioning systems, where risk-hedging firms receive preferential treatment in allocation, could incentivize stable participation and long-term investment in low-carbon technologies. Additionally, introducing longer-term trading horizons or mechanisms like carbon price floors or ceilings could reduce the speculative behavior that leads to short-term price volatility, while encouraging firms to make more balanced decisions in both carbon trading and technology investments.

CRedit authorship contribution statement

Mei Wang: Supervision, Formal analysis, Investigation, Conceptualization, Writing – review & editing, Funding acquisition. **Songyuan Liu:** Writing – original draft, Investigation, Software, Data curation, Methodology. **Jiageng Liu:** Validation, Software, Methodology. **Zhengjun Li:** Validation, Software, Methodology.

Declaration of competing interest

None.

Acknowledgements

The authors are grateful to the financial support provided by the National Natural Science Foundation of China (nos. 72243012 & 72373065 & 72104107).

Appendix A. Data sources and parameter setting

This study comprehensively captures the multidimensional heterogeneities of coal-fired power enterprises across technical, economic, and behavioral dimensions through the integration of empirical data, field surveys, and stochastic simulations, ensuring that the model accurately reflects the differentiated decision-making behavior of firms in real-world markets. To operationalize agent heterogeneity, we integrate field survey data from 100 coal-fired plants across five provinces (Shandong, Liaoning, Hebei, Tianjin, and Inner Mongolia), supplemented by Monte Carlo simulations to assign technical and economic parameters specific to coal-fired generation. Surveyed parameters include coal consumption rates, auxiliary power ratios, carbon emission intensities, O&M costs, and annual output levels. For each parameter, we fit probability distributions using plant-level operational reports (2018–2022) and industry benchmarks (e.g., China Thermal Power Efficiency Standards), as detailed in [Table A1](#).

Table A1
Parameter distributions of the model.

Parameter	Distribution type	Mean	Std.Dev	Source
Coal Consumption (g/kWh)	Lognormal	272	53	Plant operational reports (2020–2023)
Auxiliary Power Ratio (%)	Beta ($\alpha = 2, \beta = 4.5$)	6.5	1.5	CEC Thermal Plant Guidelines
Carbon Intensity (g/kWh)	Truncated Normal	810	50	CEIC Database
O&M Cost (10 k CNY)	Lognormal	2850	220	Survey data
Initial Allowance (Million ton)	Uniform	485.33	–	Provincial ETS Allocation Rules

Notes: *Auxiliary power ratios* reflect coal-specific infrastructure: older plants (pre-2010) average 8–9 %, while newer plants (post-2015) achieve 6–7 %. *Carbon intensity* incorporates coal type (e.g., bituminous vs. lignite) and combustion efficiency. The parameters of coal consumption are sampled from plant-specific distributions, with adjustments for technology vintage (e.g., ± 5 % for ultra-supercritical units). Monte Carlo sampling adjusts for regional fuel price disparities (e.g., Inner Mongolia: 550 CNY/ton vs. Shandong: 670 CNY/ton).

The construction of technical heterogeneity is grounded in historical operational data (2018–2022) from power plants, with key parameters such as coal consumption rates and auxiliary power ratios calibrated using lognormal and beta distributions. For instance, coal consumption parameters account for technological generational differences—ultra-supercritical units (commissioned post-2015) exhibit 5 % lower coal consumption compared to subcritical units (pre-2010)—while dynamically adjusting for regional disparities through thermal efficiency standards. Inner Mongolian plants, benefiting from abundant coal resources and lower transportation costs, exhibit an average auxiliary power ratio of 6.2 %, whereas Shandong plants, constrained by grid peaking demands, show a higher ratio of 7.1 %. Additionally, the model incorporates stochastic technology upgrade mechanisms, where profitability and policy stringency jointly influence adoption probabilities. For example, units with annual profits exceeding 100 million CNY have a 25 % annual probability of upgrading to ultra-supercritical technology, dynamically simulating heterogeneous pathways of technological advancement.

Behavioral heterogeneity is operationalized through a dual mechanism of trading strategy classification and dynamic reinforcement learning adaptation. Drawing on high-frequency carbon market analysis and interprovincial allowance trading logs, firms are categorized into three behavioral types: speculative (30 %), arbitrage-seeking (30 %), and risk-averse (40 %). The decision-making criteria of the firms are detailed in Section 2.4. Crucially, behavioral typologies are not static labels but evolve dynamically through multi-agent reinforcement learning (MARL): speculative agents prioritize short-term profit maximization via reward functions emphasizing quarterly gains, while risk-averse agents optimize long-term stability by penalizing allowance deficits in their reward structures, thereby simulating adaptive strategy adjustments under evolving policy conditions. To validate the heterogeneity design, Monte Carlo simulations confirm that the 30 %–30 %–40 % behavioral distribution optimally balances policy response diversity and market stability. Empirical validation against 2021–2022 market data reveals a Mean Absolute Percentage Error of less than 10 %. This precision underscores the robustness of heterogeneity parameterization and provides a reliable computational foundation for policy scenario analysis.

In addition to these firm-specific heterogeneities, the model incorporates market-level data to ensure realistic agent behavior. Agents are trained using real price data from the China national carbon market, the electricity market, and broader energy markets, exposing them to actual market signals and dynamics.

Besides, the initial system parameters of the model are detailed in [Table A2](#). The parameters related to carbon trading in the model mainly refer to the relevant policy documents of China’s carbon market. In addition, the total initial carbon quota is set to the total carbon emissions of the base year to ensure that the initial state of the model is consistent with the actual situation.

Table A2
System parameters of the model.

Parameter	Setting	Declare
N	878	Number of firms
T	365	Number of days in the performance period
K	3600	Number of seconds in each day
γ	2 %	Emission reduction targets
p^f	610	Average fuel price (CNY)
ξ	1.5	Penalty factor for excess emissions
δ_i	5 %	The proportion of the auction quota
φ	100 %	The reduction ratio of excess emissions
TW_i	40 %	The technical expectation weight
PB_i	4	The expected payback period

Appendix B. Technical parameters and supplementary analysis

The model incorporates 150 low-carbon technologies tailored to coal-fired power generation, categorized into supplementary (e.g., boiler efficiency retrofits, turbine upgrades) and innovative types (e.g., carbon capture and storage, biomass co-firing). Key parameters, including energy-saving cost (ECS_j), investment cost (CI_j), operation cost (MC_j) are calibrated using China's *Thermal Power Retrofit Guidelines* and pilot project data. Supplementary technologies are characterized by lower investment costs and moderate emission reductions, while innovative technologies exhibit higher costs but greater emission reduction potential. For instance, supplementary technologies (e.g., ID 1–22) dominate early adoption phases, with initial deployments exceeding 500 units for cost-effective retrofits ($AC < 20$ CNY/tCO₂). Besides, part of innovative technologies (e.g., ID 23, 53, 62) exhibit slower diffusion due to high AC (>30 CNY/tCO₂) and limited initial applications (≤ 100 units), aligning with real-world adoption barriers.

Initial adoption levels at the start of the compliance period follow a Pareto distribution, and firm scale modulates adoption capacity. The parameters related to low-carbon technologies are set based on existing research and policy guidelines to ensure a reasonable representation of real-world conditions. Sensitivity analyses further test parameter robustness, such as varying CI_j ($\pm 20\%$) and learning rates for innovative technologies. This granular representation ensures the model captures the heterogeneous economic and technical constraints shaping firms' adoption behavior, providing actionable insights for policymakers to tailor support mechanisms.

We categorize the low-carbon technologies adopted by power generation firms into two main types: supplementary (75 numbers) and innovative (75 numbers). Fig. B1 illustrates the adoption of these technological types by firms throughout the simulation period. Notably, the adoption volume of supplementary technologies is approximately three times that of innovative technologies, indicating a stronger inclination among firms towards low-carbon technologies with lower emission reduction costs for their technological upgrades.

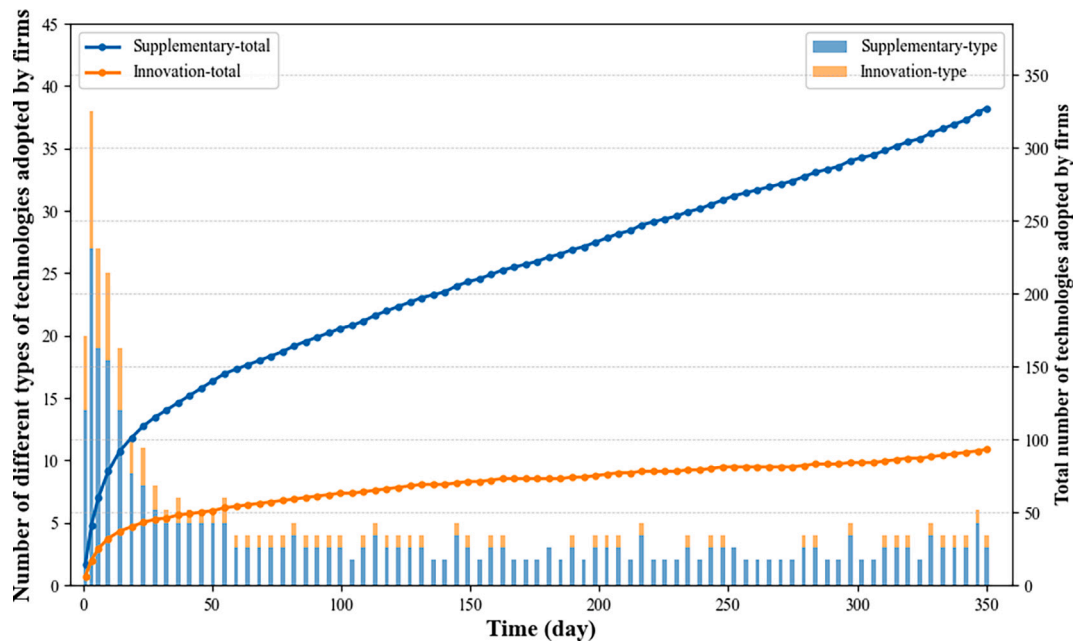


Fig. B1. The adoption of supplementary and innovative low-carbon technologies by power firms.

This preference can be attributed to several factors. First, the Technology Acceptance Model suggests that an individual's acceptance of new technologies is closely related to their perceived usefulness, ease of use, and social influence. Supplementary technologies typically feature lower investment costs and higher technological maturity, making them more attractive to small and medium-sized power enterprises. Second, supplementary technologies often achieve significant emission reductions within a shorter timeframe, enabling firms to respond quickly to policy requirements. As shown in Supplementary dataset, the average investment cost and emission reduction cost of supplementary technologies are lower than those of innovative technologies. Furthermore, policy orientation also significantly influences firms' technology choices. Government-set emission reduction targets may initially drive firms to adopt technologies with faster emission reduction effects to achieve rapid compliance.

However, from a long-term perspective, innovative technologies are crucial for achieving deep decarbonization and promoting industrial transformation and upgrading. Innovative technologies possess higher emission reduction potential and can fundamentally alter firms' production methods. Currently, firms face numerous challenges in adopting innovative technologies, including high R&D costs, technological uncertainties, and talent shortages. These challenges often lead firms to be more conservative in their technology choices, favoring the adoption of more mature supplementary technologies. To accelerate the adoption of innovative low-carbon technologies by power generation firms, concerted efforts from the government, enterprises, and research institutions are needed to foster a favorable innovation environment and promote the R&D, dissemination, and application of these advanced technologies.

Appendix C. Algorithm training

This study employs the MADDPG algorithm as the core framework, selected for its ability to model continuous action spaces—essential for simulating firms' sequential decisions in production, pricing, and allowance trading. Unlike independent Q-learning approaches, MADDPG utilizes a centralized critic to coordinate multi-agent policies, effectively addressing interdependencies among heterogeneous market participants. The framework adopts a centralized training with decentralized execution paradigm, enabling agents to leverage collective experiences for critic updates while maintaining autonomy in decision-making, thereby balancing computational efficiency with realistic agent interactions. To further enhance the

critic's capability, an attention mechanism is integrated into the critic network. This mechanism dynamically assigns weights to interactions between agents based on their state-action relevance, allowing the critic to prioritize influential agents and improve value function estimation.

The actor and critic networks are structured with three hidden layers (128, 256, and 128 neurons) using ReLU activation, optimized via the Adam algorithm with learning rates of 0.01 (actor) and 0.005 (critic). A batch size of 1024 and an experience replay buffer of 1 million transitions ensure stable training by decorrelating sampled data. Target networks are updated through a soft update strategy ($\tau = 0.005$) to mitigate value estimation divergence, while a discount factor ($\gamma = 0.95$) emphasizes long-term rewards. To optimize hyperparameters, a grid search was conducted across learning rates (0.001–0.05), batch sizes (64–1024), buffer capacities (500 k–2 M), and discount factors (0.9–0.99). The final configuration prioritized the highest average reward over 10,000 training episodes.

Two initialization strategies were tested to evaluate robustness: a random strategy with uniformly sampled actions and a history-driven strategy pre-trained on 2019–2023 China carbon market data. While the history-driven strategy accelerated convergence (1200 episodes vs. 2000 episodes for random initialization), both strategies converged to consistent steady-state outcomes, including equilibrium carbon prices (85–90 CNY/ton) and large firms' dominance in high-cost technology adoption. Training stability was ensured through gradient clipping ($[-1.0, 1.0]$), randomized experience replay, and convergence criteria based on a 1000-episode moving average reward. Training halted when reward improvements fell below 0.1 % for 500 consecutive episodes, ensuring reliable and reproducible results across multiple runs. This comprehensive approach balances computational efficiency with empirical realism, providing a robust foundation for simulating complex interactions in electricity-carbon markets.

Data availability

The data utilized in this study are partially sourced from open databases (Global Power Database) and purchasable databases (Wind Database). Some supplementary data involve commercial and non-public data, which are available from the corresponding author upon reasonable request.

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